Manojkumar I. Gupta

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Research Profile

Experimental fluid mechanics researcher working at the interface of multiphase turbulence and cloud microphysics. Current doctoral work investigates how coherent vortical structures and turbulent shear redistribute inertial droplets, generate preferential concentration, modify relative motion, and influence clustering, collision, and coalescence. The experimental framework combines canonical air vortex rings and turbulent boundary layers with high-speed shadowgraphy, time-resolved PIV/PTV, and MATLAB-based statistical analysis. The broader objective is to connect vortex-scale droplet dynamics with turbulence-scale statistics relevant to warm-cloud bottleneck physics and collision-coalescence pathways.

Education

Degree	Institute / University	CGPA	Duration
PhD in Applied Mechanics	Indian Institute of Technology Delhi	9.30	Aug. 2021 – Present
BTech in Mechanical Engineering	Charotar University of Science and Technology, Gujarat	9.16	2016 – 2020

Publication

Jha, N. K., Gupta, M. I., & Govardhan, R. N. (2026). Interaction of a bubble with vortical structures in a horizontal turbulent channel flow. *Physics of Fluids*, 38(1).
DOI: 10.1063/5.0309012

Conference Presentations and Poster

- **2nd European Fluid Dynamics Conference (EFDC2)**, University College Dublin, Dublin, Ireland, 26–29 August 2025. Contribution: *Interaction of droplets with vortical structures: to understand atmospheric clouds.*
- **12th International and 52nd National Conference on Fluid Mechanics and Fluid Power (FMFP 2025)**, Nirma University, Gujarat, India, 19–21 December 2025. Paper ID: FMFP2025-MPF-274. Contribution: *Interaction of cloud-like droplets with the vortical structure.*
- **PMRF Annual Symposium 2026**, IIT Guwahati, India, 6–7 March 2026. Poster presented: *Interaction of polydisperse droplets with an air vortex ring: towards understanding atmospheric cloud.*

Research Experience and Projects

PhD Research, IIT Delhi *Aug. 2021 – Present*
Supervisor: Prof. Narsing Kumar Jha, FE2B Fluids Lab
Experimental study of droplet interaction with coherent vortical structures and turbulent boundary layers, with emphasis on clustering, preferential concentration, collision-coalescence, and cloud-microphysics-relevant transport dynamics.

Evaluation of Wind Loads from Wind Tunnel Tests, IIT Delhi *Jan. 2025 – Feb. 2025*
With Prof. Srinivas Veeravalli and Prof. Narsing Kumar Jha
Worked on the evaluation of wind loads using an aeroelastic scaled model of the 150 m high single-steel flue RCC chimney for 500 MW TPS, Ukai, Gujarat (GSECL), in the Boundary Layer Wind Tunnel of IIT Delhi.

BTech Project *July 2019 – April 2020*
Charotar University of Science and Technology
Design and fabrication of a multi-purpose mechanical machine.

Teaching and Academic Service

Teaching Assistantship, IIT Delhi

- Fluid Mechanics; Fluid Dynamics; Engineering Mechanics; Engineering Thermodynamics; Experimental Methods for Fluids and Solids; Advanced Experiments in Fluids and Solids

PMRF Teaching Duty

- Intermediate Physics, V. V. Inter College, Tochhigarh, Aligarh
- Engineering Thermodynamics, Mechanical Vibrations, Fluid Mechanics, Machine Design Lab, Machine Drawing Lab, Fluid Systems, Hydraulics and Pneumatics, Refrigeration and Air Conditioning, and instruction related to limits, fits, and tolerances at G. B. Pant Institute of Technology, Delhi

Selected Coursework

- Applied Fluid Dynamics
- Physics of the Atmosphere
- Modeling of Transport Processes
- Cloud Physics
- Engineering Mathematics and Computation
- Hydrodynamic Stability
- Physics of Turbulent Flows
- Computational Fluid Dynamics
- Experimental Methods

Workshops

- Visualization of Fluid Flow and Temperature Distribution, FMFP 2023, IIT Jodhpur, 12–15 July 2023
- Practical Electronics Workshop, IIT Delhi, 3–6 October 2023
- Dispersed Multiphase Flow Fundamentals, conducted by Prof. S. Balachandar through NPTEL, 4–6 October 2024
- Particle-based Flow Diagnostics: From field measurements to 3D Holographic Imaging, conducted by Prof. Jiarong Hong, University of Minnesota, 15, 22, and 29 March 2025

Funding, Awards, and Technical Skills

Funding and Awards

- Prime Minister's Research Fellowship (PMRF) for PhD research
- ANRF funding support for research
- GATE 2021 qualified, percentile 94.81 in Mechanical Engineering
- Gujarat PGCET 2020, Rank 3 in Mechanical Engineering

Technical Skills
Experimental methods: High-speed imaging, shadowgraphy, time-resolved PIV, and PTV.
Programming and analysis: MATLAB and LabVIEW.
Platforms: Linux and Windows.